姓名：

学号：

学院：

年级：

上海科技大学

2022-2023 学年第一学期期末考试卷

开课单位：物质科学与技术学院

授课教师：杨晓瑜，高得伟，黄焕明

考试科目：**有机化学基础**

课程序号：CHEM1322.01, CHEM1322.02, CHEM1322.03

考生须知：

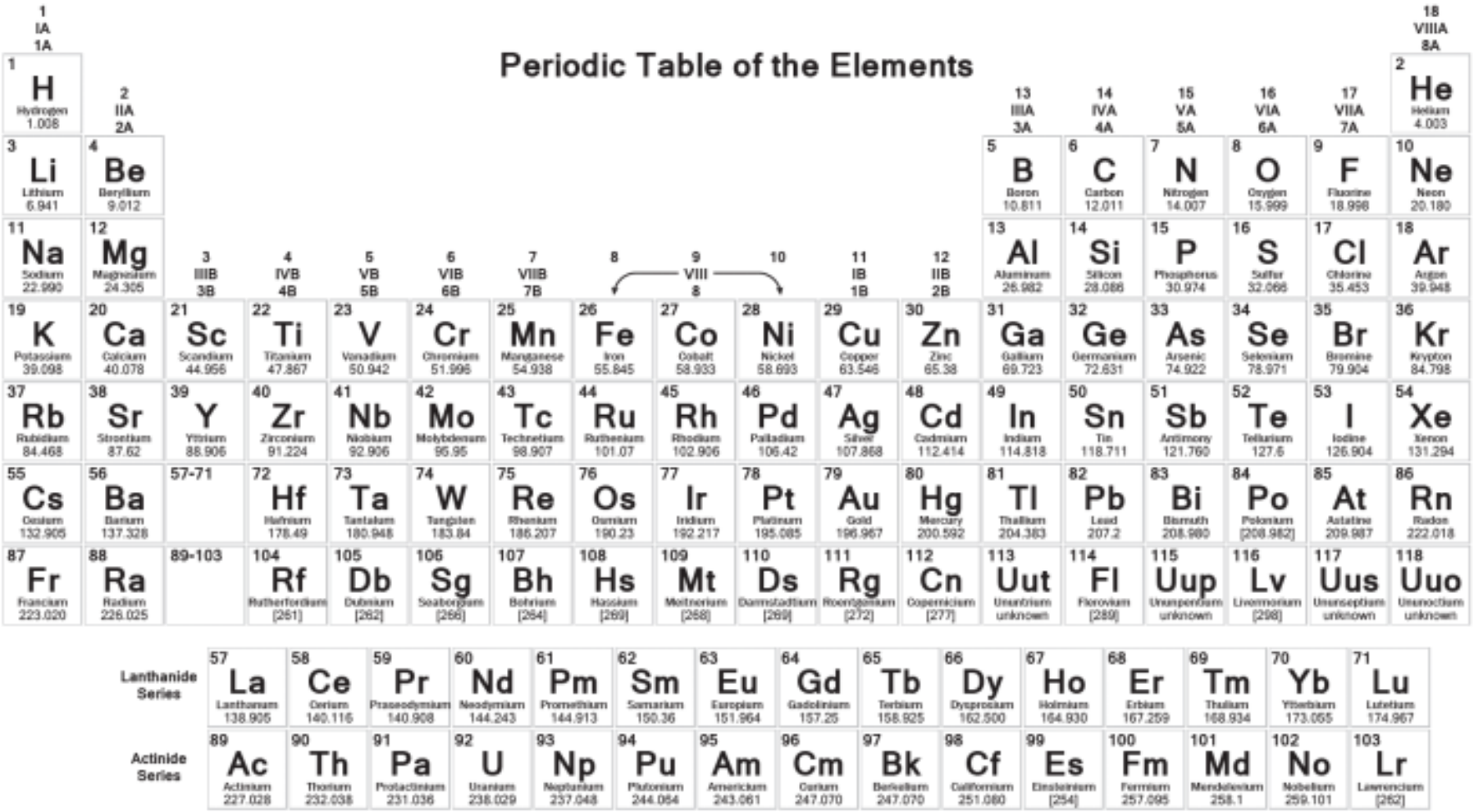
1. 请严格遵守考场纪律，禁止任何形式的作弊行为。
2. 参加闭卷考试的考生，除携带必要考试用具外，书籍、笔记、掌上电脑和其他电子设备等物品一律按要求放在指定位置。
3. 本试卷共 8 道大题。总计满分 100 分。
4. 答题可任选中文或英文。

考试成绩录入表：

题目	1	2	3	4	5	6	7	8	**总分**
计分									
复核									

评卷人签名： **复核人签名：**

日期： **日期：**


Periodic Table of the Elements
1 IA 1A
2 IIA 2A
3 IIIB 3B
4 IVB 4B
5 VB 5B
6 VIB 6B
7 VIIB 7B
8 9 VIII 10 8
11 IB 1B
12 IIB 2B
13 IIIA 3A
14 IVA 4A
15 VA 5A
16 VIA 6A
17 VIIA 7A
18 VIIIA 8A
1 H Hydrogen 1.008
2 He Helium 4.003
3 Li Lithium 6.941
4 Be Beryllium 9.012
5 B Boron 10.811
6 C Carbon 12.011
7 N Nitrogen 14.007
8 O Oxygen 15.999
9 F Fluorine 18.998
10 Ne Neon 20.180
11 Na Sodium 22.990
12 Mg Magnesium 24.305
13 Al Aluminum 26.982
14 Si Silicon 28.086
15 P Phosphorus 30.974
16 S Sulfur 32.066
17 Cl Chlorine 35.453
18 Ar Argon 39.948
19 K Potassium 39.098
20 Ca Calcium 40.078
21 Sc Scandium 44.956
22 Ti Titanium 47.867
23 V Vanadium 50.942
24 Cr Chromium 51.996
25 Mn Manganese 54.938
26 Fe Iron 55.845
27 Co Cobalt 58.933
28 Ni Nickel 58.693
29 Cu Copper 63.546
30 Zn Zinc 65.38
31 Ga Gallium 69.723
32 Ge Germanium 72.631
33 As Arsenic 74.922
34 Se Selenium 78.971
35 Br Bromine 79.904
36 Kr Krypton 84.798
37 Rb Rubidium 84.468
38 Sr Strontium 87.62
39 Y Yttrium 88.906
40 Zr Zirconium 91.224
41 Nb Niobium 92.906
42 Mo Molybdenum 95.95
43 Tc Technetium 98.907
44 Ru Ruthenium 101.07
45 Rh Rhodium 102.906
46 Pd Palladium 106.42
47 Ag Silver 107.868
48 Cd Cadmium 112.414
49 In Indium 114.818
50 Sn Tin 118.711
51 Sb Antimony 121.760
52 Te Tellurium 127.6
53 I Iodine 126.904
54 Xe Xenon 131.294
55 Cs Cesium 132.905
56 Ba Barium 137.328
57-71
72 Hf Hafnium 178.49
73 Ta Tantalum 180.948
74 W Tungsten 183.84
75 Re Rhenium 186.207
76 Os Osmium 190.23
77 Ir Iridium 192.217
78 Pt Platinum 195.085
79 Au Gold 196.967
80 Hg Mercury 200.592
81 Tl Thallium 204.383
82 Pb Lead 207.2
83 Bi Bismuth 208.980
84 Po Polonium [208.982]
85 At Astatine 209.987
86 Rn Radon 222.018
87 Fr Francium 223.020
88 Ra Radium 226.025
89-103
104 Rf Rutherfordium [261]
105 Db Dubnium [262]
106 Sg Seaborgium [266]
107 Bh Bohrium [264]
108 Hs Hassium [269]
109 Mt Meitnerium [268]
110 Ds Darmstadtium [269]
111 Rg Roentgenium [272]
112 Cn Copernicium [277]
113 Uut Ununtrium unknown
114 Fl Flerovium [289]
115 Uup Ununpentium unknown
116 Lv Livermorium [298]
117 Uus Ununseptium unknown
118 Uuo Ununoctium unknown
Lanthanide Series
57 La Lanthanum 138.905
58 Ce Cerium 140.116
59 Pr Praseodymium 140.908
60 Nd Neodymium 144.243
61 Pm Promethium 144.913
62 Sm Samarium 150.36
63 Eu Europium 151.964
64 Gd Gadolinium 157.25
65 Tb Terbium 158.925
66 Dy Dysprosium 162.500
67 Ho Holmium 164.930
68 Er Erbium 167.259
69 Tm Thulium 168.934
70 Yb Ytterbium 173.055
71 Lu Lutetium 174.967
Actinide Series
89 Ac Actinium 227.028
90 Th Thorium 232.038
91 Pa Protactinium 231.036
92 U Uranium 238.029
93 Np Neptunium 237.048
94 Pu Plutonium 244.064
95 Am Americium 243.061
96 Cm Curium 247.070
97 Bk Berkelium 247.070
98 Cf Californium 251.080
99 Es Einsteinium [254]
100 Fm Fermium 257.095
101 Md Mendelevium 258.1
102 No Nobelium 259.101
103 Lr Lawrencium [262]

1. (10 pts) Draw the structures for the following corresponding names

a) 2-Cyclohexenecarboxylic acid

b) 2-fluoro-5-methylphenol

c) (*R*)- 3-hydroxybutanoic acid

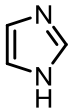
d) N,N-Dimethylbenzamide

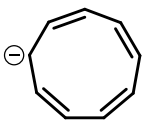
e) (*R*)-2-Phenyl-1-propanamine

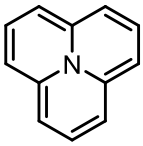
2. (8 pts) Please answer the following questions as shown below.

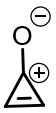
(a) Which ones of the following compounds are aromatic?

a) 

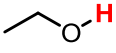
b) 

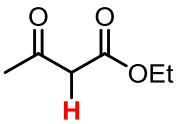
c) 

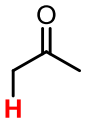
d) 

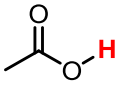
e) 

(b) Arrange the order of the acidity of the highlighted proton in the following compounds.

(a) 

(b) 

(c) 

(d) 

3. (9 pts)

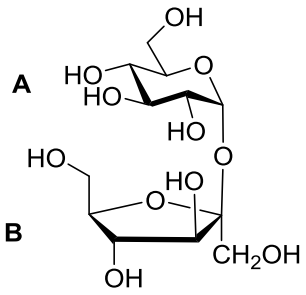
a) Compound A has the formula $\mathbf{C_6H_{12}O}$. The ^{13}C NMR has an obvious signal at δ 190-200 ppm. IR spectrum shows a strong absorption peak at 1724 cm^{-1}. The ^{1}H NMR spectrum data is given below: 0.92 (t, 3H), 1.10 (d, 3H), 1.68 (m, 2H), 2.21 (s, 3H), 2.42 (m, 1H).

1) Calculate the **index of hydrogen deficiency** for compound **A**.
2) Deduce the structure of compound **A**.

b) Compound B has the formula $\mathbf{C_{10}H_{12}O_2}$. IR spectrum shows a strong absorption peak at 1724 cm^{-1} and a moderate absorption peak at 3010 cm^{-1}. The ^{1}H NMR spectrum data is given below: 1.3 (t, 3H), 2.4 (q, 2H), 5.1 (s, 2H), 7.3 (m, 5H).

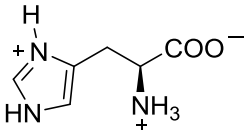
1) Calculate the **index of hydrogen deficiency** for compound **B**.
2) Deduce the structure of compound **B**.
3) How many ^{13}C peaks could be observed in the ^{13}C NMR spectrum of this compound.

4. (9 pts) The structure of a disaccharide (二糖) is shown below.



1) Indicate the configuration of the linkage between these two saccharides (for saccharide **A** and **B**, respectively).
2) Is this disaccharide a reducing sugar?
3) Draw the **Fisher project** of each monosaccharide (单糖) **A** and **B**.
4) Show how the monosaccharide (单糖) **B** was converted into aldonic acids (醛糖酸)。

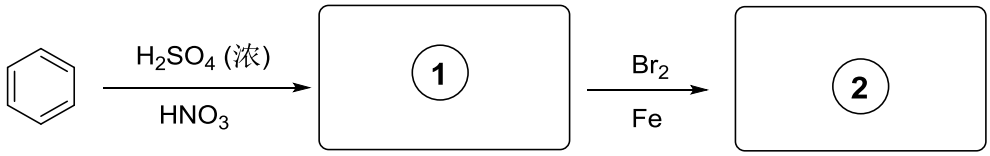
5. (6 pts) Histidine (组氨酸) has the pK_a values of 1.77, 9.18, 6.10.

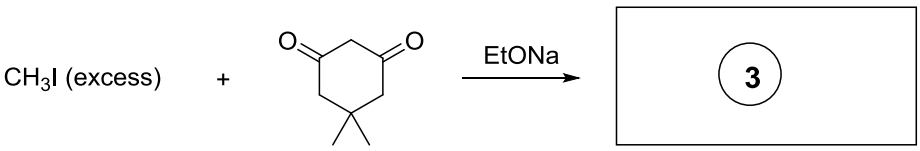


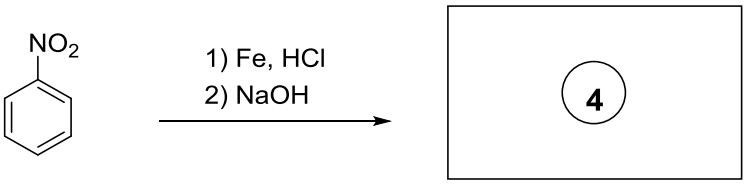
Histidine (H)

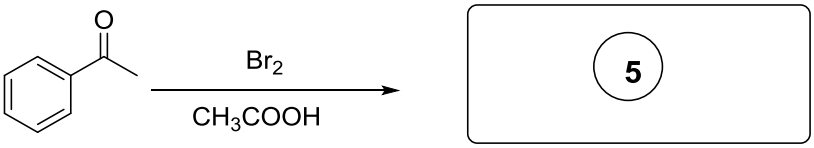
1) Draw the resonance structure of the imidazole (咪唑) group of Histidine.
2) Assign the each pK_a value of Histidine.
3) Calculate the PI (Isoelectric point, 等电点) of Histidine.
4) What's the net charge of Histidine at pH = 9?

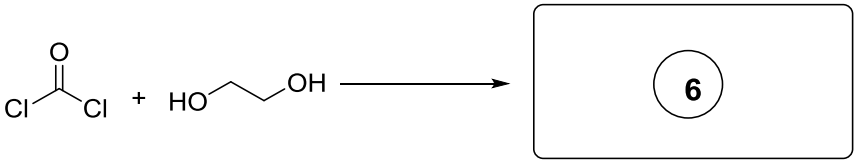
6. (28 pts) Fill the blanket to finish the following reactions (**more than one step is possible**)

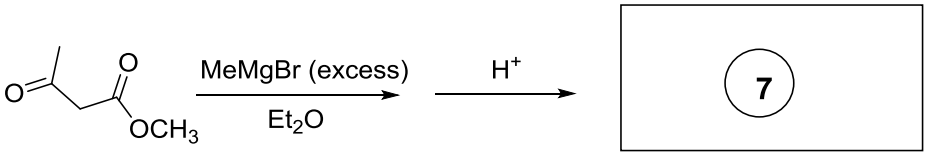


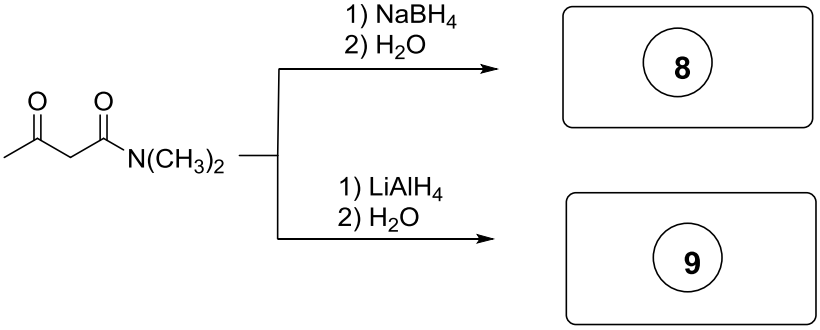


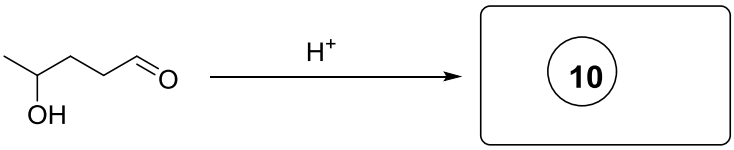


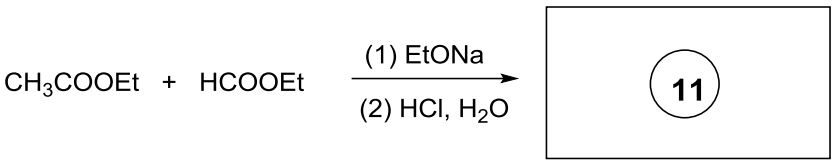


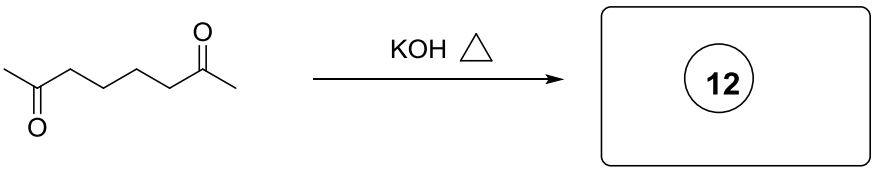


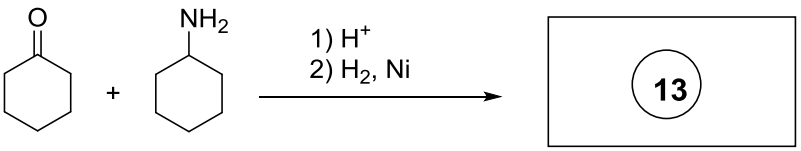


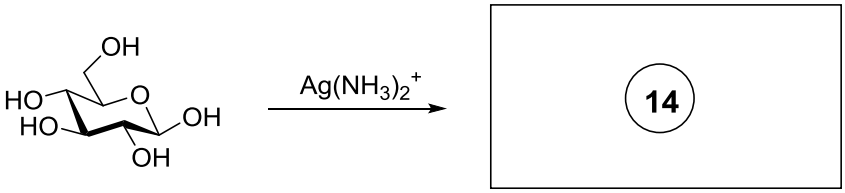

1) NaBH4
2) H2O
O
O
N(CH3)2
1) LiAlH4
2) H2O
8
9


H+
O
OH
10


CH3COOEt + HCOOEt
(1) EtONa
(2) HCl, H2O
11

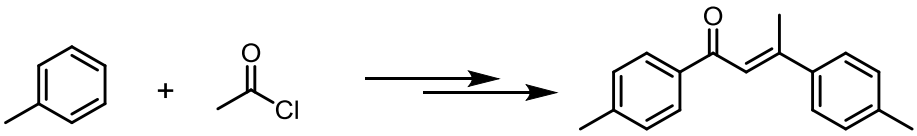

O
O
KOH △
12


O
NH2
+
1) H+
2) H2, Ni
13

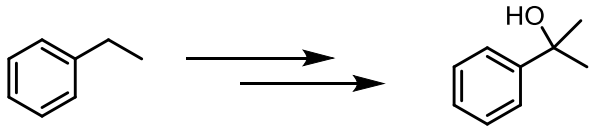

OH
HO
O
OH
HO
OH
Ag(NH3)2+
14

7. (**15 pts**) Propose a synthetic route to the following product from the corresponding starting material (you can use some other reagents if necessary)

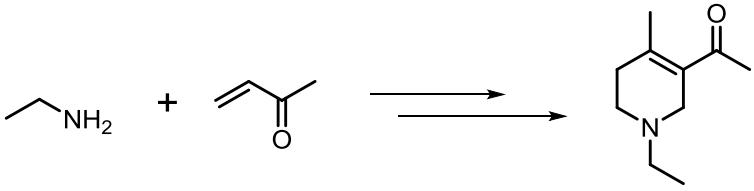
(1)



(2)

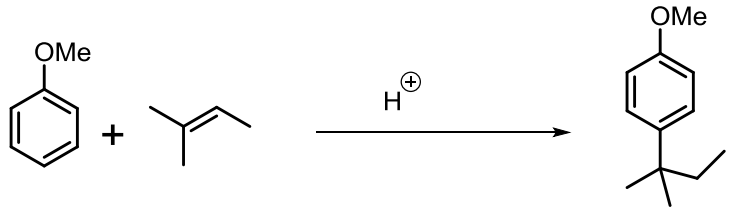


(3)



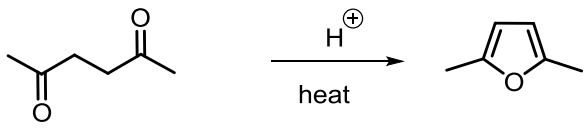
8. (**15 pts**) Draw the detailed mechanism for the following transformations.

(1)



And explain the origin of the regioselectivity.

(2)



(3)

